

Computer Networks and Data Communication

Module 6

CS 212

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Module wise CONTENTS:

- **Module 1 - Introduction:** An overview of networks, the Internet, services and protocols, RFCs and standards of Networking, Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.
- **Module 2 - Physical Layer:** Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).
- **Module 3 - Data Link Layer:** Overview; sliding window protocols - Stop-&-Wait; Go-Back-N, Selective Repeat; Error detection and correction; Hamming Code, LRC, VRC, CRC, Checksum, Round Trip Time, Latency (Delay); Medium Access Sublayer: medium access control (MAC) Addresses; Switching techniques - circuit, message & packet; Random Access Protocols - Pure ALOHA, Slotted ALOHA, CSMA / CD, CSMA / CA; Controlled Access Protocols – FDMA, TDMA, CDMA; Channelization Protocols – Reservation, Polling, Token Passing; Ethernet (802.3 IEEE standard), Wireless Links, Wireless LAN - WiFi (802.11).
- **Module 4 - Network Layer:** Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing Protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).
- **Module 5 - Transport Layer:** Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. **Application Layer:** Introduction – features & functions; application layer protocols; Web and HTTP, Email, P2P network, its types & applications.
- **Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.**

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- Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.

Network Optimization:

- Network optimization refers to the set of techniques and strategies used to improve the **performance, efficiency, reliability, and speed** of a computer network. The goal is to ensure data moves through the network as quickly and smoothly as possible while using resources efficiently.

Key Goals of Network Optimization

- **Reduce latency** – minimize delays in data transmission.
- **Increase throughput** – maximize the amount of data transferred over time.
- **Balance network load** – avoid congestion and uneven traffic distribution.
- **Ensure reliability** – maintain stable, fault-tolerant network operations.
- **Improve resource usage** – optimize bandwidth, routing, and hardware use.

Network Optimization:

Techniques Used in Network Optimization

1. Traffic Analysis & Monitoring

- Monitoring tools (e.g., NetFlow, Wireshark) help identify bottlenecks and unusual traffic patterns.

2. Load Balancing

- Distributes traffic across multiple servers or paths to prevent overload.

3. QoS (Quality of Service)

- Prioritizes critical applications (video conferencing, VoIP) over less urgent traffic.

4. Caching

- Stores frequently accessed data closer to users to reduce retrieval time.

Network Optimization:

Techniques Used in Network Optimization

5. Compression

- Compresses data payloads to speed up transmission.

6. Routing Optimization

- Uses algorithms to find the fastest and least-congested paths for data (e.g., OSPF, BGP tuning).

7. Bandwidth Management

- Limits or allocates bandwidth to prevent congestion from specific apps.

8. Network Design Optimization

- Improving topology—like using redundant links, segmenting networks (VLANs), or placing servers strategically.

Network Optimization:

■ **Tools Commonly Used**

- Cisco DNA Center
- SolarWinds Network Performance Monitor
- Nagios
- Wireshark
- NetFlow analyzers

■ **Benefits of Network Optimization**

- Faster data transfer
- Better application performance
- Improved user experience
- Lower operational costs
- Enhanced network stability and security

Mathematical Models

- Mathematical models are formal frameworks used to **analyze, predict, and optimize** the behavior and performance of computer networks. They help engineers understand how networks behave under various conditions and guide the design of efficient and reliable communication systems.
- **Mathematical models allow us to:**
 - Predict network performance before deployment
 - Analyze congestion, delay, throughput, and packet loss
 - Optimize routing and resource allocation
 - Improve reliability and fault tolerance
 - Evaluate protocols and algorithms analytically

Major Types of Mathematical Models in Computer Networks

1. Graph Theory Models

- Used to represent network topology.
- Nodes → routers, switches, hosts
- Edges → communication links

Applications:

- Shortest path routing (Dijkstra, Bellman-Ford)
- Network connectivity & reliability
- Minimum spanning tree (MST)
- Load balancing and path optimization

2. Queueing Theory Models

- Used to analyze packet arrival, waiting time, and service time.
- Common models:
- **M/M/1 queue** → single server, exponential arrival & service
- **M/M/c queue** → multiple servers
- **M/G/1 queue** → general service distribution

Applications:

- Delay estimation
- Buffer sizing
- Congestion analysis
- Performance of routers/switches

3. Probability & Stochastic Models

- Used for modeling randomness in:
 - Traffic arrival
 - Packet loss
 - Radio signal variability
 - User behavior

■ Examples:

- Markov chains
- Poisson processes
- Random walks

4. Optimization Models

- Used for maximizing/minimizing network performance metrics.

■ Optimized metrics:

- Bandwidth usage
- Energy consumption
- Routing cost
- Network reliability

■ Techniques include:

- Linear programming
- Integer programming

Mathematical Models

Major Types of Mathematical Models in Computer Networks

5. Simulation Models

- Mathematical + algorithmic modeling using tools like:
 - NS-3
 - OPNET
 - OMNeT++
- **Used when analytical models are too complex.**
- **6. Traffic Models**
 - Used to characterize and predict network traffic patterns.
 - Deterministic traffic models
 - Stochastic traffic models
 - Self-similar (long-range dependence) models

■ **Applications:**

- Bandwidth planning
- QoS engineering
- Congestion control

Mathematical Models

- **Benefits of Mathematical Models**
 - Better understanding of network behavior
 - Improved design and planning
 - Predictable performance
 - Helps troubleshoot bottlenecks
 - Reduces overall cost and complexity

Introduction Advance Computer Networks

- Advanced Computer Networks focus on the **design, analysis, implementation, and optimization** of large-scale, high-performance communication systems. These networks go beyond basic connectivity, supporting **high-speed data transfer, intelligent routing, automation, security, and global-scale communication**.
- Advanced Computer Networks are modern communication systems that incorporate **sophisticated technologies**, such as:
 - High-speed broadband and fiber-optic systems
 - Wireless and mobile networks (4G/5G/6G)
 - Software-Defined Networking (SDN)
 - Network Function Virtualization (NFV)
 - Cloud and edge networking
 - IoT and sensor networks
 - AI-driven network management
- These networks are designed to handle **massive traffic loads, ensure reliability, minimize latency, and allow automation**.

Introduction Advance Computer Networks

Key Features of Advanced Computer Networks

1. High Performance

- Supports gigabit to terabit data rates with low latency and efficient routing.

2. Scalability

- Networks expand easily to accommodate millions of connected devices.

3. Security

- Advanced encryption, intrusion detection, firewalls, and authentication.

4. Automation

- AI/ML and SDN enable automatic configuration and optimization.

5. Reliability & Fault Tolerance

- Redundant paths, dynamic routing, and load balancing ensure uninterrupted service.

Introduction Advance Computer Networks

Core Concepts in Advanced Networks

1. Network Architecture

- Layered architecture, protocol stacks, and network topologies.

2. Routing & Switching

- Advanced algorithms (e.g., OSPF, BGP), MPLS, and traffic engineering.

3. Wireless & Mobile Communication

- Radio frequency handling, mobility management, handoff techniques.

4. Cloud & Edge Networks

- Distributed computing and content delivery networks (CDNs).

5. Virtualization

- SDN and NFV separate network functions from hardware, enabling flexibility.

MANET, WSN, PCN, SDN, IoT

MANET, WSN, PCN, SDN, IoT

- These are key advanced networking paradigms used in modern communication systems. Each represents a different type of network architecture designed for specific use-cases such as mobility, sensing, virtualization, or large-scale connectivity.

1. MANET (Mobile Ad-hoc Network)

- A MANET is a **self-configuring, infrastructure-less wireless network** where each node acts as both a host and a router.

Key Features

- Nodes are mobile
- Dynamic topology
- No centralized control
- Multi-hop communication

Applications

- Disaster recovery
- Military communication
- Vehicular networks (VANET)
- Temporary event networks

MANET, WSN, PCN, SDN, IoT

2. WSN (Wireless Sensor Network)

- WSNs consist of **sensor nodes** that collect, transmit, and process data about environmental conditions.

Characteristics

- Low power, limited memory
- Dense deployment
- Multi-hop routing
- Energy-efficient communication

Applications

- Agriculture monitoring
- Industrial automation
- Health monitoring
- Smart homes and smart cities

MANET, WSN, PCN, SDN, IoT

3. PCN (Pervasive Computing Network)

- PCNs are networks enabling **ubiquitous computing**, where devices embedded in the environment communicate seamlessly.

Key Ideas

- Context-aware communication
- Invisible computing
- Integration of heterogeneous devices

Applications

- Smart environments
- Wearable devices
- Intelligent transportation
- Ambient assisted living

MANET, WSN, PCN, SDN, IoT

4. SDN (Software-Defined Networking)

- SDN separates the **control plane** (decision making) from the **data plane** (packet forwarding), enabling **centralized network management**.

Important Features

- Programmable network
- Centralized controller
- Dynamic traffic engineering
- Suitable for cloud & data centers

Advantages

- Flexibility
- Reduced complexity
- Rapid innovation (via controllers like OpenFlow)

MANET, WSN, PCN, SDN, IoT

5. IoT (Internet of Things)

- IoT is a network of **interconnected physical devices** that collect and exchange data.

Characteristics

- Massive number of devices
- Heterogeneous communication
- Cloud/edge connectivity
- Real-time monitoring

Applications

- Smart homes
- Industrial IoT (IIoT)
- Healthcare IoT
- Smart grid & smart cities

MANET, WSN, PCN, SDN, IoT

Acronym	Full Form	Key Feature	Typical Use-Case
MANET	Mobile Ad-hoc Network	Mobile, infrastructure-less	Military, disaster recovery
WSN	Wireless Sensor Network	Sensing & monitoring	Agriculture, environment
PCN	Pervasive Computing Network	Ubiquitous devices	Smart environments
SDN	Software-Defined Networking	Programmable, centralized	Data centers, cloud
IoT	Internet of Things	Connected physical devices	Smart homes, industries

Computer Networks and Data Communication

